

The 3rd Meeting of the Special Interest Group in Non-Equilibrium Molecular Dynamics

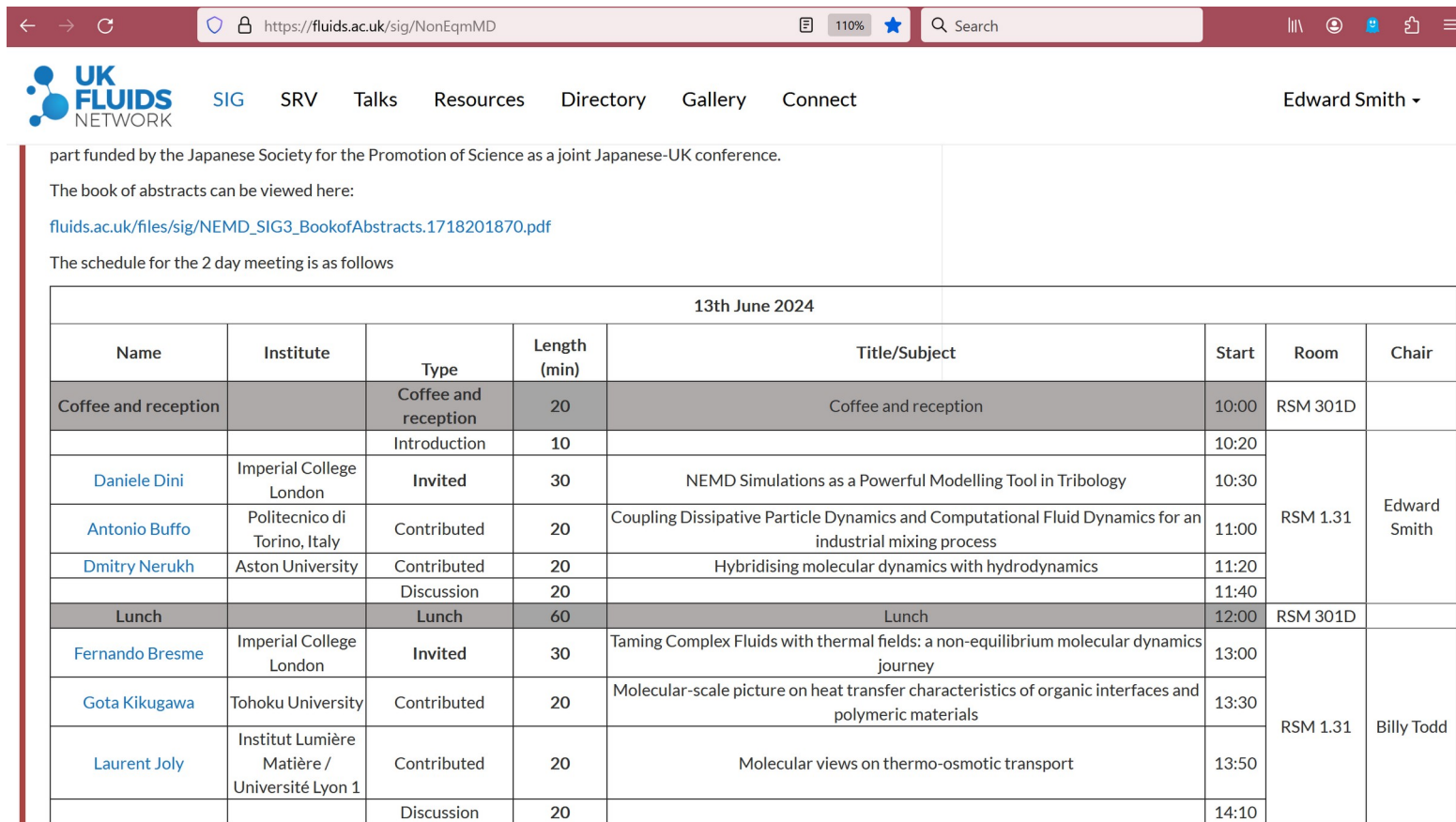
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13/06/2024

Imperial College London

Plan for Today

Plan on the website: <https://fluids.ac.uk/sig/NonEqmMD>



part funded by the Japanese Society for the Promotion of Science as a joint Japanese-UK conference.

The book of abstracts can be viewed here:
fluids.ac.uk/files/sig/NEMD_SIG3_BookofAbstracts.1718201870.pdf

The schedule for the 2 day meeting is as follows

13th June 2024								
Name	Institute	Type	Length (min)	Title/Subject	Start	Room	Chair	
Coffee and reception		Coffee and reception	20	Coffee and reception	10:00	RSM 301D		
		Introduction	10		10:20			
Daniele Dini	Imperial College London	Invited	30	NEMD Simulations as a Powerful Modelling Tool in Tribology	10:30			
Antonio Buffo	Politecnico di Torino, Italy	Contributed	20	Coupling Dissipative Particle Dynamics and Computational Fluid Dynamics for an industrial mixing process	11:00	RSM 1.31	Edward Smith	
Dmitry Nerukh	Aston University	Contributed	20	Hybridising molecular dynamics with hydrodynamics	11:20			
		Discussion	20		11:40			
Lunch		Lunch	60	Lunch	12:00	RSM 301D		
Fernando Bresme	Imperial College London	Invited	30	Taming Complex Fluids with thermal fields: a non-equilibrium molecular dynamics journey	13:00			
Gota Kikugawa	Tohoku University	Contributed	20	Molecular-scale picture on heat transfer characteristics of organic interfaces and polymeric materials	13:30	RSM 1.31	Billy Todd	
Laurent Joly	Institut Lumière Matière / Université Lyon 1	Contributed	20	Molecular views on thermo-osmotic transport	13:50			
		Discussion	20		14:10			

Also downstairs, on walls and in the abstract (scan QR code)

Plan for Today

- Sessions are short: one Invited talk (30 min), two contributed (20 min) followed by discussion time (20 min)
 - Discussions can be Faraday style questions for all 3 speakers
 - Then moving to discuss wider issues, e.g. linking NEMD to industrial problems, the importance of AI, future of NEMD, etc
 - Please feel free to suggestion topics to discuss
- Session are also roughly grouped by theme to aid discussion (apologies if your talk is not a perfect fit)
 - Multiscale modelling, thermal flows, liquid-vapour interfaces, droplets/wetting, boiling, wall slip, tribology and alternative methods
 - Discussion in these sessions ideally follow these but not essential

What is the UK Fluids Network (UKFN)?

- EPSRC Funded (EP/N032861/1) in 2016 (ended 29th Feb 2020)
 - Aims to keep the UK leading in fluid mechanics
 - To engage as a group with industry
 - To build leadership within the community
 - To facilitating inter-disciplinary research
- Funded 1st Special Interest Groups (SIGs) which paid for the first meeting at Brunel University (mostly UK academics)
- 2nd meeting was online (during covid) and more international
 - Moving from Australia in the morning to China, India, Europe and ending in the US by afternoon.
- 3rd meeting here with travel funded by JSPS, ARC and kindly hosted by Imperial for free (with food and organisation supported by Professor Daniele Dini)

NEMD SIG Subject and Scope

- Non-equilibrium molecular dynamics (NEMD) is the study of fundamental fluid flow using MD
- The non-equilibrium betrays the subject's origins, away from thermodynamic equilibrium due to
 - Wall motion, temperature and pressure gradients
 - Interfaces, liquids-solid and liquid-vapour, boiling and tribology.
- This is **Fluid Dynamics**, traditionally modelled by continuum differential equations
 - Molecular models allow a different way to simulate these problems
 - NEMD has a strong statical mechanics theoretical underpinning
 - Fundamental expertise has the potential to revolutionise nano-scale fluid dynamics – an area of growing importance for industry.

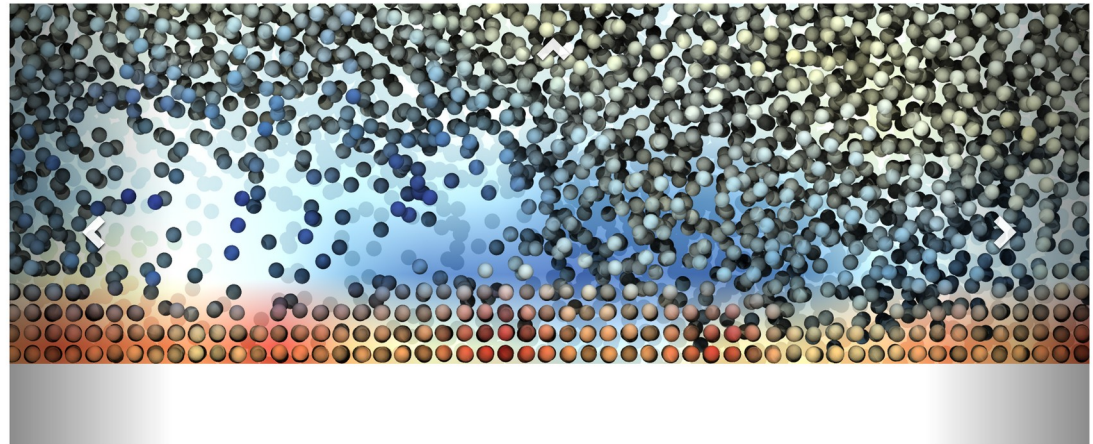
Our SIG in Non-Equilibrium Molecular Dynamics (NEMD)

<https://fluids.ac.uk/sig/NonEqmMD> www.fluids.ac.uk/sig/NonEqmMD



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Edward Smith ▾



Non-equilibrium Molecular Dynamics (NEW)

Description

Non-equilibrium molecular dynamics (NEMD) is the study of fundamental fluid flow using molecular simulation - 'non-equilibrium' because the system is driven away from thermodynamic equilibrium by wall motion, temperature and pressure gradients or

Tweets

@UKFluidsNetwork

 UK Fluids Network Retweeted



- We have a mailing list of all SIG members and past attendees
- I will add everyone here to a mailing list for future events (please let me know if you'd prefer I didn't)

UK-Japan joint conference to be held in Japan from April 2025 to March 2026



1. Osaka

- Excl. late Jul.-Aug. (too hot)
- Near historical cities (Kyoto, Nara)
- Expo will be held (13 Apr.-13 Oct.)
- Easy access

2. Tokyo

- Excl. late Jul.-Aug. (too hot)
- Easiest access

3. Fukuoka

- Excl. late Jul.-Aug. (too hot)
- Hot spring (2 hour trip)

4. Sendai

- Possible in summer
- Hot spring (1 hour trip)

5. Sapporo

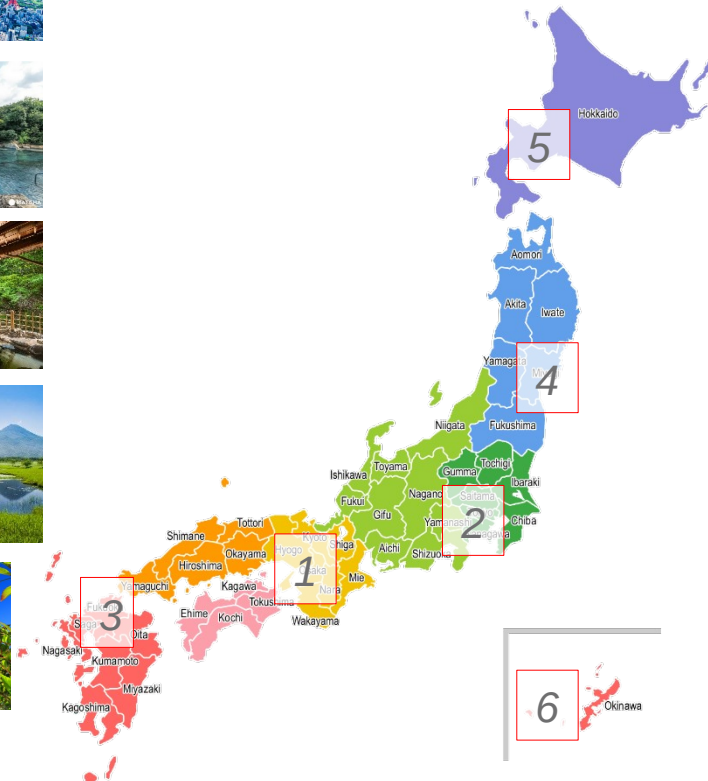
- Possible in summer, cold in winter
- Nature & hot spring

6. Okinawa

- Fixed date (middle May 2025) with domestic heat transfer conf.
- Tropical island



London-Tokyo: ~14 hours
Sydney-Tokyo: ~10 hours



Thank You

- Rohit and James for helping to organise
- Daniele, Muhammad and Andi
- Yasutaka and our Japanese visitors
- Our invited speakers and chairs
- Everyone here for making the journey